

FIELD OPERATIONS NETWORK ARCHITECTURE

PROJECT ORION

One network. Every mission. A standardized digital backbone for Civil Air Patrol field operations.

5

KITS

Modular by design. Core, Trailer, Base, Fleet, Air.

1

NETWORK

One Wi-Fi credential. One dial plan. Every mission.

0

LICENSE FEES

No per-seat costs, no SaaS lock-in on the management plane.

When a mission starts, the radios work. The data network does not, and someone scrambles to fix it. **Every time.** ORION ends the scramble: a federated mesh of standardized kits that connect every trailer, mission base, vehicle, aircraft, and field team under one logical fabric.

THE PREMISE

Civil Air Patrol's voice and radio communications are mature. The data side has not **kept pace.**

Personal hotspots fail in rural areas. Wi-Fi coverage is fragmented. There is no shared file system, no consistent dashboards, and limited interoperability with partner agencies. ORION is an architecture of standardized kits, built on commercial UniFi networking and Starlink backhaul, that turns mission connectivity from an improvised setup into a Wing-wide capability.

THE FIVE KITS

001 — 005

001	ORION-Core	Wing HQ controller and primary anchor for the architecture.	● IN PROGRESS
002	ORION-Trailer	Rapid-deploy mobile core in the comms trailer.	● FIELD TESTED
003	ORION-Base	Mission base extension for staff in a building or hangar.	● PROPOSED
004	ORION-Fleet	Every Wing vehicle becomes a network node with TAK tracking.	● PROPOSED
005	ORION-Air	In-flight connectivity for live aerial imagery and TAK position.	● CONCEPT

PATH FORWARD

Funded grassroots. Built to last.

Phase 1 stands up the architecture in Michigan Wing as the proof of concept. From there, results build the case for National engagement and a sustainable CAP-USAF line item. To support a build, fund a phase, or collaborate across Wings, reach out.

POINT OF CONTACT

SMSgt Luke Bunge, CAP

Director of Information Technology
Michigan Wing Civil Air Patrol

luke.bunge@miwg.cap.gov

USAF Cyber Warfare Warrant Officer